

EXPERIMENT 15. Probabilistic Context-Free Grammar (PCFG) Parsing

Aim: To implement probabilistic context-free grammar parsing for a sentence using Python.

Program:

```
import nltk

from nltk import PCFG

from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [0.5]
NP -> 'John' [0.5]
VP -> V NP [1.0]
Det -> 'the' [1.0]
N -> 'dog' [1.0]
V -> 'sees' [1.0]
""")

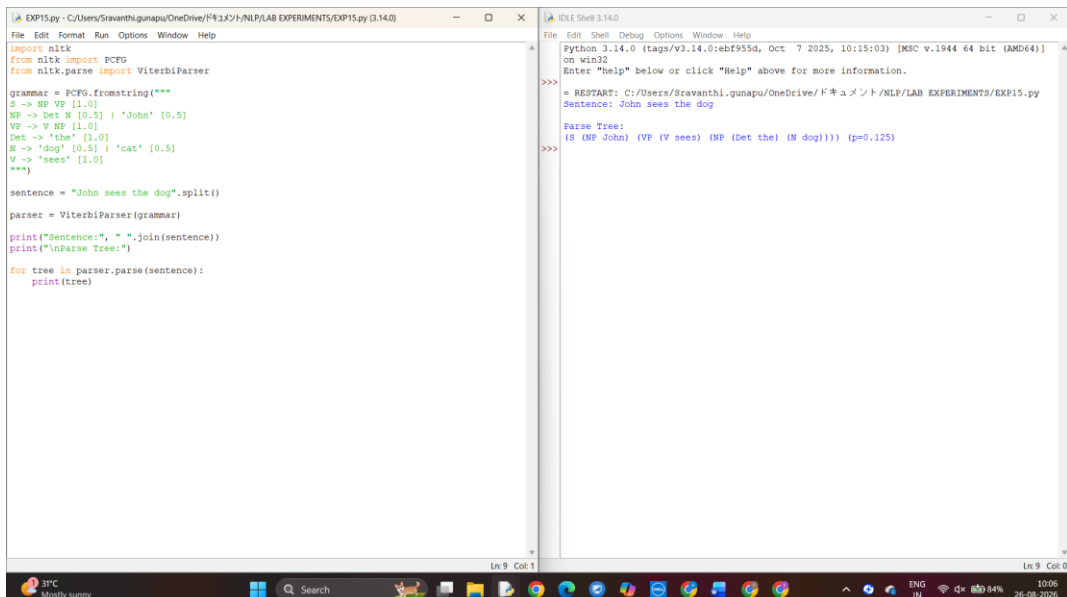
parser = ViterbiParser(grammar)

sentence = "John sees the dog".split()

for tree in parser.parse(sentence):

    print(tree)
```

Output:



```
EXP15.py - C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP15.py (3.14.0)
File Edit Format Run Options Window Help
import nltk
from nltk import PCFG
from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [0.5] | 'John' [0.5]
VP -> V NP [1.0]
Det -> 'the' [1.0]
N -> 'dog' [0.5] | 'cat' [0.5]
V -> 'sees' [1.0]
""")

sentence = "John sees the dog".split()
parser = ViterbiParser(grammar)

print("Sentence:", " ".join(sentence))
print("\nParse Tree:")

for tree in parser.parse(sentence):
    print(tree)

Python Shell 2.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP15.py
Sentence: John sees the dog
Parse Tree:
(S (NP John) (VP (V sees) (NP (Det the) (N dog)))) (p=0.125)
>>>
```